

Building Management System

CSIR Campus · Building 53

Continuous monitoring and automatic control of HVAC plant and temperature-critical medicine storage, built by Spartcon on the Tridium Niagara 4 platform with an on-site JACE 8000 controller.

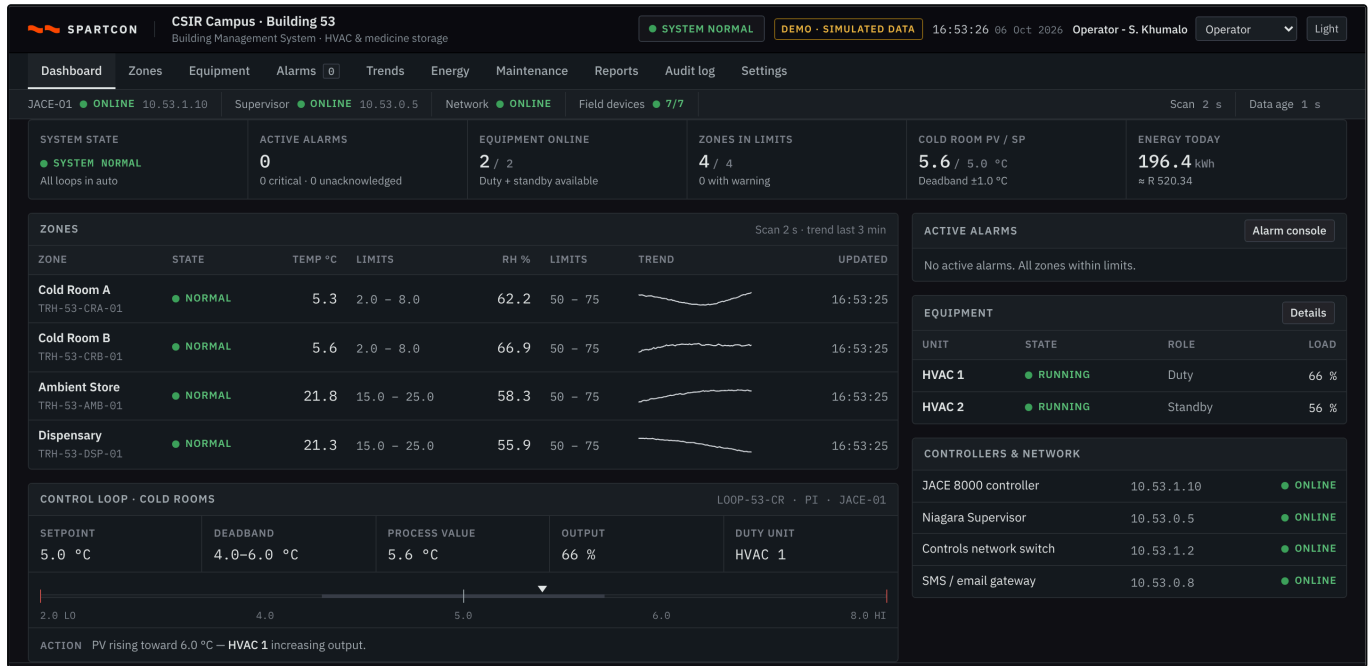


Fig. 1 — Operator dashboard. System state, every zone against its allowed limits, the cold-room control loop, active alarms and equipment status on a single screen.

MONITOR

Temperature and humidity in Cold Rooms A and B (2–8 °C), the Ambient Store and the Dispensary (15–25 °C), scanned every 2 seconds and logged for trending and compliance reporting.

CONTROL

A duty / standby HVAC pair is controlled automatically from the warmest cold-room probe. The duty unit ramps up as rooms warm toward 6 °C and down toward 4 °C; standby shares load in normal operation.

PROTECT

Pre-alarms and alarms with SMS and email notification, automatic escalation if nobody responds, and automatic failover to the standby unit if the duty unit fails or stops communicating.

4 monitored zones	2 s scan rate	2 HVAC units, duty / standby	30 s escalation if unacknowledged	3 user roles with audit trail
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RESILIENCE

Automatic failover and alarm escalation

If the duty HVAC unit fails, the standby unit takes over within seconds and the right people are told, in order, until someone takes responsibility. Medicine storage stays in range throughout.

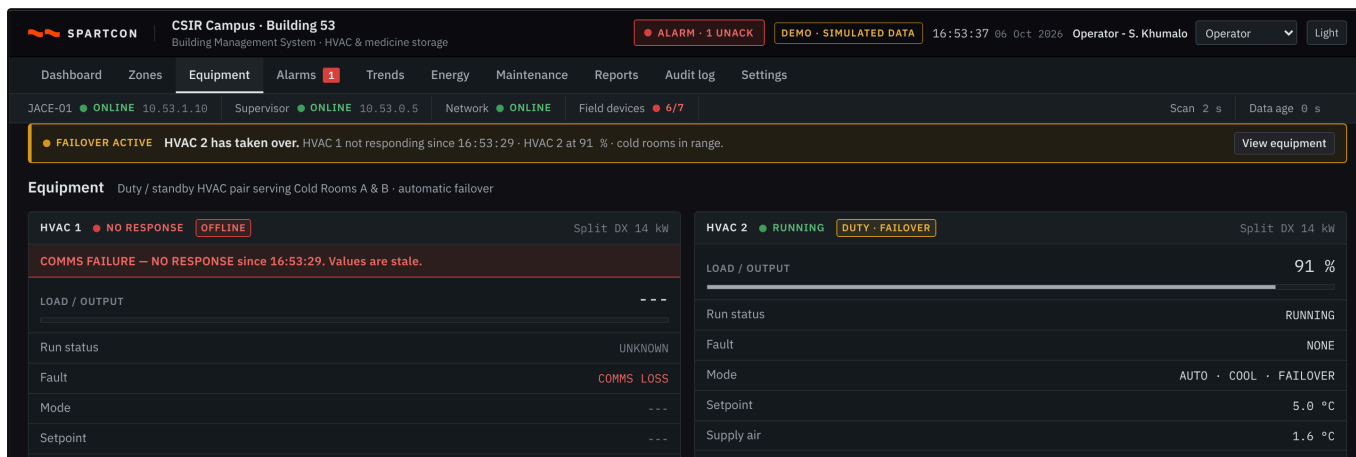


Fig. 2 — Equipment during a failover. HVAC 1 has stopped responding and its values are flagged as stale; HVAC 2 has started automatically and is carrying the full cold-room load.

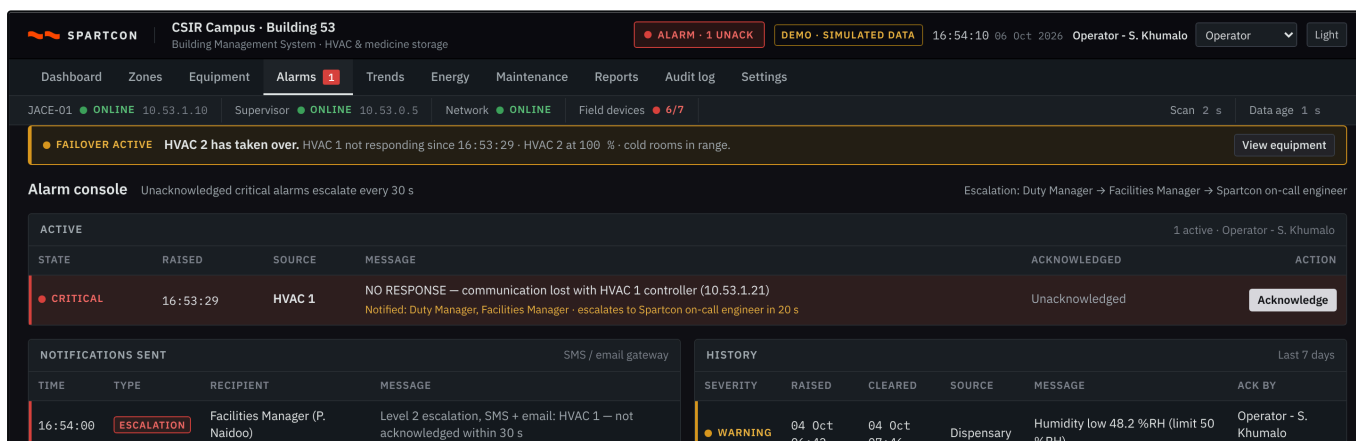


Fig. 3 — Alarm console. The active critical alarm, the escalation countdown, every SMS / email sent and the 7-day alarm history with who acknowledged each alarm and when.

TIME	EVENT	SYSTEM RESPONSE
T + 0 s	● Duty unit HVAC 1 stops responding	Critical alarm raised; SMS and email to the Duty Manager
T + 2 s	● Automatic failover	HVAC 2 started and ramped to 100 %; escalation alert to the Spartcon on-call engineer; failover banner shown to operators
T + 30 s	● Alarm still unacknowledged	Escalated to the Facilities Manager; further escalation every 30 s
On ack	● Operator acknowledges	Escalation stops; operator name and time stamped on the alarm and in the audit log
On restore	● HVAC 1 back online	Normal duty / standby load sharing resumes; alarm closed to history; event kept as failover-test evidence

COMPLIANCE

Trends, excursions and weekly reporting

Every reading is logged, so any period outside the allowed limits is visible, counted and reported. Screens shown are from the working concept demonstration; all values are simulated.

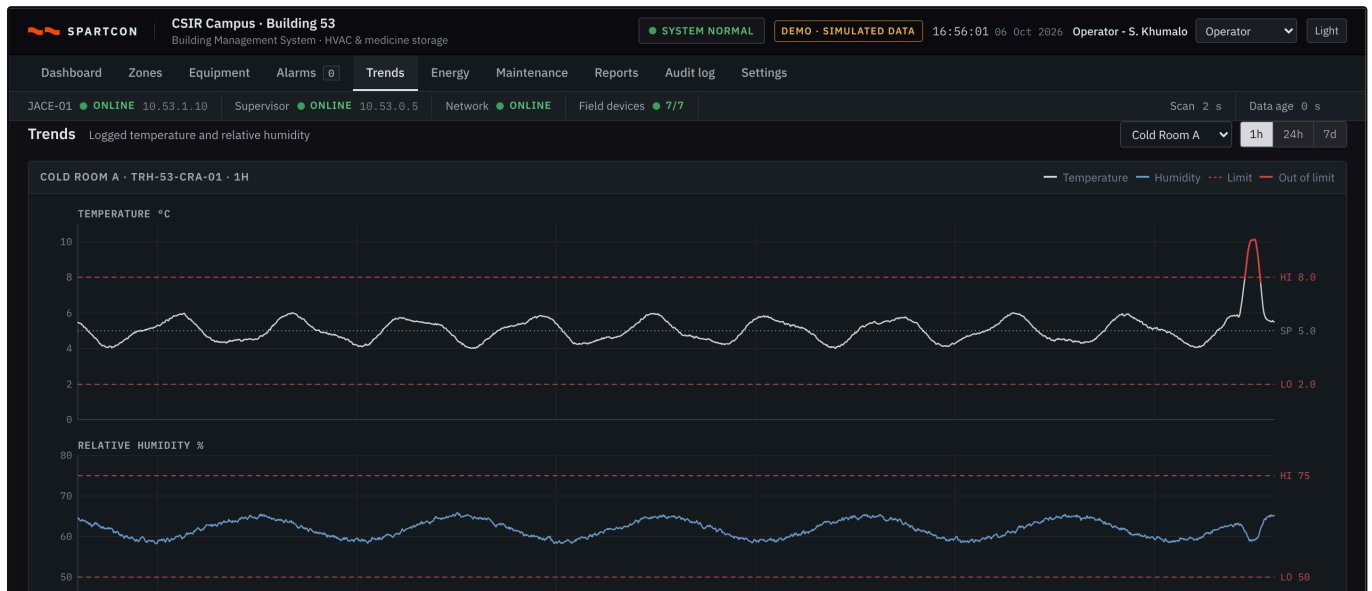


Fig. 4 — Cold Room A trend, last hour. Temperature and humidity against the allowed limits. The simulated door-open excursion above 8 °C is highlighted in red, followed by recovery once the alarm was acknowledged and the door closed.

The screenshot shows the 'Reports' page with a table titled 'WEEKLY ENVIRONMENTAL REPORT'. The table displays compliance data for various zones over a specific period. The data is as follows:

ZONE	LIMITS	MIN °C	AVG °C	MAX °C	AVG RH %	TEMP EXC.	RH EXC.	IN LIMITS
Cold Room A	2-8 °C	4.0	5.0	10.1	62.0	1	0	99.99 %
Cold Room B	2-8 °C	4.2	5.2	8.4	64.1	1	0	99.40 %
Ambient Store	15-25 °C	18.9	21.0	23.1	56.0	0	0	100.00 %
Dispensary	15-25 °C	20.2	22.2	24.0	57.7	0	1	100.00 %

Fig. 5 — Weekly environmental report. Minimum, average and maximum per zone, excursion counts and time within limits, exportable as CSV for audits.

ARCHITECTURE & SECURITY

- JACE 8000 controller on site; Niagara Supervisor for history and reporting
- Field devices on BACnet/IP and Modbus RTU
- Remote access via VPN only — controllers are not exposed to the internet
- Role-based access: Viewer, Operator, Administrator
- Full audit log of alarms, acknowledgements and setpoint changes

DELIVERED AND SUPPORTED BY SPARTCON

- Design, installation and commissioning of controls and sensors
- 6-monthly preventive service of HVAC plant
- Annual SANAS-traceable probe calibration
- Monthly duty / standby failover testing
- 24/7 on-call engineer in the escalation chain